

PRAJJWAL RAUT

+91-7498531368

✉ prajjwalraut226@gmail.com

in linkedin.com/in/prajjwal-raut

github.com/prajjwalraut



Professional Summary

Machine Learning, Deep Learning and Generative AI specialist with hands-on experience building end-to-end ML/DL pipelines and LLM-powered applications using Python, TensorFlow, PyTorch, OpenAI APIs, and HuggingFace Transformers. Proven expertise in supervised/unsupervised learning, deep learning architectures, NLP, prompt engineering, and RAG systems. Skilled in model experimentation, data preprocessing, feature engineering. Strong problem-solver with quick learning aptitude and collaborative mindset.

Education

Vishwakarma Institute of Information Technology, Pune

2023 – 2027

B.Tech in Computer Engineering CGPA: 8.52

12th: 78.17% (2023) 10th: 88.20% (2021)

Skills

Languages: Python, Java, SQL

AI/ML: Supervised Learning (Regression, Classification), Unsupervised Learning (Clustering, PCA), Data Preprocessing, Feature Engineering, Model Training & Evaluation, Hyperparameter Tuning, Cross-Validation, Neural Networks, CNN, ANN, RNN, NLP

Generative AI: LangChain, LangGraph, Large Language Models (GPT, BERT), Prompt Engineering, Transformers Architecture, Embeddings (Word2Vec, Sentence Transformers), Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, Pinecone), Hallucination Mitigation, AI Agents

Tools: Git, Jupyter Notebook, Google Colab, VS Code, Docker (basic), REST APIs

Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks

Projects

Multi-PDF Retrieval-Augmented Generation System

- Developed a Retrieval-Augmented Generation (RAG) based research assistant capable of answering contextual queries across multiple PDF documents.
- Implemented document chunking, semantic retrieval, and vector similarity search using LangChain and ChromaDB.
- Integrated Groq LLM API with retrieval pipelines to generate grounded responses with source-aware context injection.
- Built a Streamlit-based interactive interface supporting multi-document querying, conversational memory, and citation-aware responses.
- **Tech Stack:** Python, LangChain, ChromaDB, Streamlit, Groq API, HuggingFace Embeddings, RAG

Multi-Agent Placement Readiness Evaluator using LangGraph & RAG

- Developed a multi-agent AI evaluation system to assess student placement readiness across technical skills, aptitude, communication, and domain knowledge.
- Designed an agent workflow using **LangGraph** with specialized agents for assessment, response evaluation, and personalized report generation.
- Implemented adaptive question evaluation and structured scoring using **LLM-based reasoning with LangChain**.
- Built a **RAG pipeline** over placement preparation resources to generate context-aware improvement recommendations and learning roadmaps.
- **Tech Stack:** Python, LangGraph, LangChain, RAG, LLMs, Vector Database, HuggingFace, Generative AI

Silkworm & Mulberry Leaf Disease Detection using Deep Learning

- Designed and trained a custom CNN architecture achieving 89% classification accuracy on multi-class disease detection dataset.
- Built an end-to-end pipeline including **image preprocessing, normalization, and data augmentation**.
- Used **ReLU, batch normalization, and dropout** to improve convergence and reduce overfitting.
- Trained using **Adam optimizer** and evaluated using **F1-score** and confusion matrix.
- **Tech Stack:** Python, TensorFlow, NumPy, Matplotlib